Community Structure Guided Network for Hyperspectral Image Classification

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***Abstract*— Recently, the hypergraph convolutional network (HGCN) has attracted increasing attention in hyperspectral image (HSI) classification. Compared to graph convolutional networks, HGCN has a stronger ability to mine nonlinear high-order correlations. However, the problems of intraclass variability and interclass similarity exist due to the effects of light, environment, and sensor bias, resulting in insufficient reliability of hypergraphs constructed by directly utilizing the original spectral features. Motivated by the observation that the land cover in HSI contains the spatial distribution semantic information of community structures, which can be used to extract deeper contextual semantic features, we propose a novel community structure guided network (CSGNet) for HSI classification. Specifically, CSGNet adopts a dual-branch architecture: the HGCN branch focuses on superpixel-level high-order feature extraction, while the convolutional neural network (CNN) branch enhances pixel-level local features. In HGCN branch, a novel reliable hypergraph construction approach is introduced, which strikes a balance between depth-first search (DFS) and breadth-first search (BFS), effectively representing different community structure features and improving the ability of edge detection. Meanwhile, kernel function mapping is used to achieve more accurate node connections and enhances classification within classes. Finally, to achieve balanced training of the HGCN and CNN branches, we add their cross-entropy loss as an auxiliary component in the backpropagation process. Experimental results demonstrate that CSGNet outperforms the state-of-the-art methods. The code will be released at https://github.com/KustTeamWQW/CSGNet.**

***Index Terms*— Contextual semantic features, hypergraph convolutional network (HGCN), hyperspectral image (HSI) classification, reliable hypergraph.**

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I. Introduction

HYPERSPECTRAL images (HSIs) typically comprise dozens or even hundreds of spectral bands, containing a significant amount of spectral–spatial features [1]. The HSI classification aims to assign each pixel to a specific class label through its spectral–spatial features. Accurate HSI classification can provide valuable and precise information support for various fields [2], [3], making it a consistently hot topic in research.

High-order correlations in HSI play a crucial role in improving classification performance, which are summarized as complex associations between multiple spectral bands (spectral high-order correlations), as well as deep relationships between pixels beyond simple spatial neighborhoods (spatial high-order correlations) in this article. Specifically, spatial high-order correlations are not only limited to direct connections between pixels (i.e., first-order neighborhoods), but also include complex connections between pixels indirectly through intermediate pixels or regions (i.e., high-order neighborhoods), for example, through similar spectral patterns, textures, or spatial distribution structures. The deep learning methods, which can automatically extract deep-level nonlinear features, have gained widespread application, and demonstrated excellent performance in HSI classification [4], [5]. Among them, convolutional neural networks (CNNs) [6], [7], [8] are one of the most widely used methods. Specifically, CNN-based methods perform convolution operations on HSI, utilizing learnable filters to extract local features like edges and textures. Through stacking multiple layers of convolutional, the model can capture more profound features, including shapes, patterns, and structures [9]. Ranging from 1D-CNN [4] to 3D-CNN [10], and from single CNN [11] to hybrid CNN [12], [13], there has been continuous improvement in capturing spectral–spatial features. The contextual deep CNN [6], which is deeper than other existing deep methods, can optimally explore local contextual information. The spectral–spatial residual network (SSRN) [14] introduces spectral–spatial residual blocks to extract discriminative features. However, CNN-based methods are designed for Euclidean data, ignoring the topological information and long-range correlations contained in HSI. Furthermore, acquiring labeled HSI samples is challenging, yet CNN-based methods demand a substantial amount of samples to achieve satisfactory classification results.

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In recent years, graph neural networks (GNNs) [15], [16], [17] have garnered significant attention in HSI classification because they can compensate for the drawbacks of CNNs. By encoding HSI into graph structures, the geometric topology and long-range correlations contained in HSI are naturally modeled. Additionally, graph convolution has been proven effective in transmitting global features [18] and can fully utilize unlabeled samples, alleviating the burden on labeled data [19]. Therefore, an increasing number of GCN-based methods have applied to HSI classification [20], [21]. The multiscale short and long-range graph convolutional network (MSLGCN) [22] generates multiscale graphs by controlling the size of superpixels and introduces the short-long graph convolution to fully explore multiscale features. Liu et al. [23] proposed a hybrid deep network called CNN-enhanced GCN (CEGCN), which utilizes CNN and GCN branches to generate complementary spectral–spatial features at pixel and superpixel levels, respectively. Subsequently, Dong et al. [24] replaced GCN with GAT and proposed the weighted feature fusion of CNN and graph attention network (WFCG), which further improved classification performance. However, the basic graph mainly focuses on pairwise vertex correlations and cannot model the multiple vertices correlations. Additionally, the basic graph employs a one-to-one feature interaction paradigm between nodes, which will generate too much redundant information.

To solve these problems, some recent studies employ hypergraphs to model complex high-order correlations [25], [26], [27], [28]. The hypergraph is the extension of simple graph, which can connect any number of vertices with a hyperedge. Inspired by the success of deep learning, Feng et al. [25] introduced deep learning to hypergraph and proposed the first hypergraph CNN (HGCN). First, HGCN utilizes hypergraphs to encode high-order data correlations, and then a two-stage hypergraph convolution is designed for data representation learning. Subsequently, a series of HGCN-based methods [29], [30], [31], [32] have been proposed to enhance the modeling capability for complex high-order correlations. Tian et al. [27] utilized a fully weighted hypergraph to model high-order correlations in aerial images and designed a hypergraph feature pyramid network (hyper-FPN) to learn nonlocal correlations in multiscale features. Ma et al. [33] proposed a feature fusion hypergraph neural network (F2HGNN), which utilizes spectral–spatial hypergraphs to represent multimodal features in HSI and iteratively updates the weights of hypergraphs to achieve optimality. Duta et al. [34] incorporate sheaves into the hypergraph Laplacian and propose the sheaf hypergraph convolutional networks (HGCNs), creating a powerful instrument for effectively modeling complex data structures. Yan et al. [35] propose a control-diffusion hypergraph dynamic system by an ordinary differential equation, which improves the stability of hypergraph neural networks.

Although these HGCN-based methods have achieved promising results, they still encounter the following limitations for HSI classification.

1) Intraclass variability and interclass similarity exist due to the effects of light, environment, and sensor bias, resulting in insufficient reliability of hypergraphs constructed by directly utilizing the original spectral features. For example, the same type of vegetation may have varying spectral reflectance features at different growth stages, in different soil types, or under varying humidity conditions. The performance of HGCN largely depends on the quality of the hypergraph. If the constructed hypergraph is unreliable, HGCN may suffer from ineffective feature aggregation due to the presence of noisy neighbors, resulting in reduced final classification accuracy.
2) Existing methods often construct hypergraphs through manually defined distance functions, such as Euclidean distance, but they often have limited capacity to model high-dimensional nonlinear spectral features of land covers. Therefore, these methods are sensitive to interclass similarity, leading to the aggregation of features from neighbors that might belong to different classes.

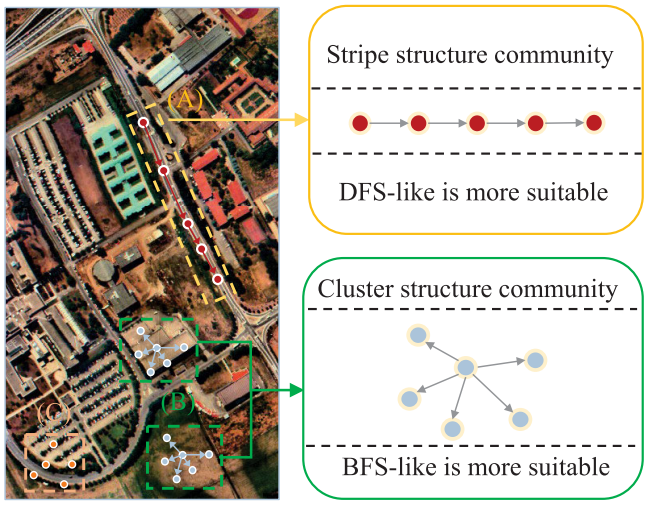


Fig. 1. Illustrations of spatial distribution semantic information of community structures contained in HSI, where region A is stripe structure community, region B is cluster structure community, and region C is scatter structure community.

In our opinion, richer semantic information can effectively alleviate this problem and improve the classification performance of HSI. Some recent studies [36], [37], [38] have also focused on the extraction of deep semantic information. Dual context network with transformer (DCN-T) [37] improves model learning by focusing on homogeneous areas. Enhanced hypergraph neural network (EHGNN) [38] utilizes key hypergraph to guide hypergraph convolution. Inspired by this, we consider the spatial distribution of land covers in HSI as a deeper level semantic information. As shown in Fig. 1, the complex spatial distribution of land covers in HSI can be considered as a mixture of cluster structure communities, stripe structure communities, and scatter structure communities. By modeling the spatial distribution structure, the model gains richer contextual information and improves its ability to detect edges. It can be observed that different community structures demand distinct processing strategies. For instance, the features of asphalt (region A) may be better mined through a depth-first search (DFS) like strategy, while the features of meadows (region B) may benefit from a breadth-first search (BFS) like strategy as shown in Fig. 1. However, DFS and

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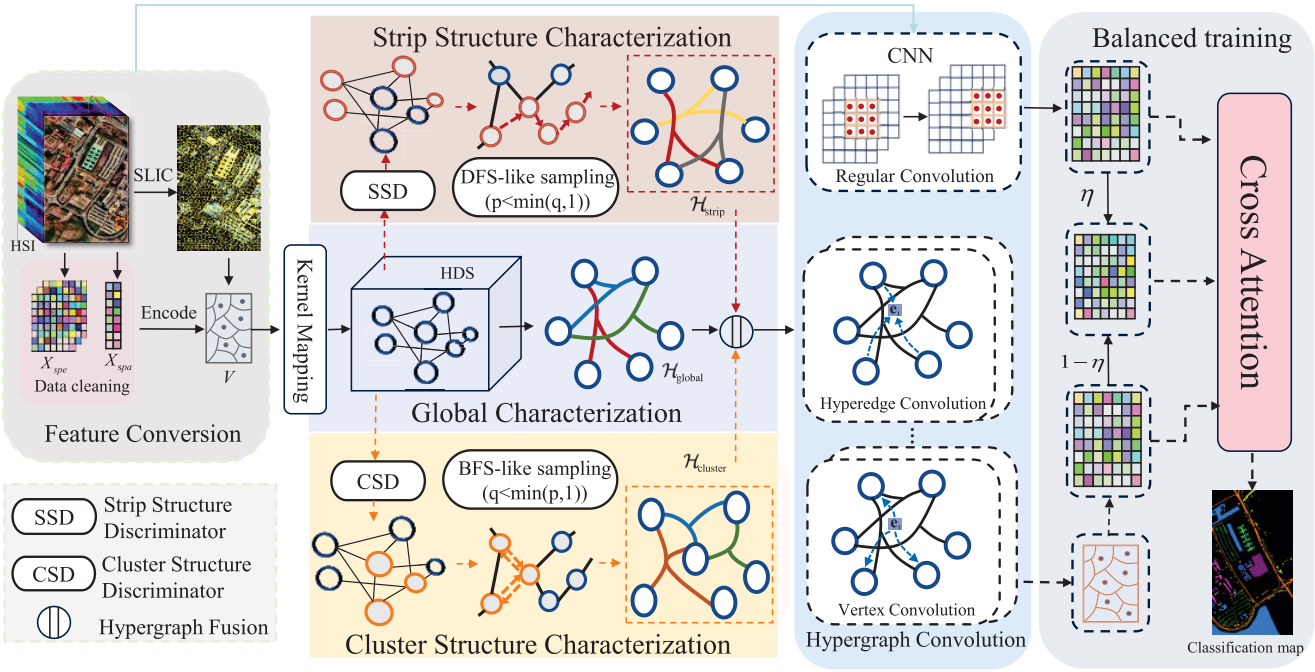


Fig. 2. Flowchart of the proposed CSGNet. Global characterization models global correlations and scatter structure communities. Strip structure characterization models stripe structure communities, while cluster structure characterization focuses on modeling cluster structure communities. Finally, HSI classification results are achieved by balancing the training of CNN and HGCN branches.

BFS represent two extreme sampling strategies. In most HSIs, distribution structures are unlikely to consistently exhibit such typical patterns. The distribution structure cannot be effectively characterized using these extreme sampling strategies. Therefore, how to effectively represent the spatial distribution information of these community structures becomes the main focus of this article.

To alleviate the above problems, a novel community structure guided network (CSGNet) is proposed for HSI classification. Specifically, the CSGNet is designed with a novel hypergraph construction approach that strikes a balance between DFS and BFS with some bias, ensuring the effective representation of different spatial distribution semantic information of community structures and improves the ability of edge detection. Meanwhile, a new approach to determine the correlations between nodes through kernel matrices is proposed, which enhances classification within classes. The main contributions of this article are as follows.

1) For HSI classification, we propose the concept of spatial distribution of community structure. Meanwhile, we present a novel reliable hypergraph construction strategy for HSI. It achieves hyperedge node sampling between DFS and BFS with some bias through a second-order biased random walk strategy, resulting in an effective characterization of community structure information and improving the reliability of hypergraph.
2) A new approach is proposed to determine the connections between nodes. It employs the kernel function to map the original spectral–spatial features into a high-dimensional feature space, thereby mitigating the limitations of manually defined distance functions when dealing with high-dimensional nonlinear spectral–spatial features and enhances classification within classes.
3) A novel CSGNet is proposed for HSI classification. CSGNet is a dual-branch structure, which has a strong ability to capture high-order correlations. Experiments on three HSI datasets demonstrate the effectiveness of CSGNet.

The rest of this article is organized as follows. In Section II, the proposed CSGNet is introduced. In Section III, experiments are described. The discussion is performed in Section IV. Finally, Section V summarizes this article.

II. Proposed CSGNet

A. Overall Network Architecture

The overall network architecture of our proposed CSGNet is shown in Fig. 2. CSGNet is a dual-branch structure, where the HGCN branch is used for superpixel high-order feature extraction, and the CNN branch is employed for pixel-level local feature extraction. The HGCN branch uses kernel function mapping to achieve more accurate node connections and leverages hypergraphs to model the spatial distribution semantic information of community structures. This enhances the reliability of the hypergraph and improves both the ability of edge detection and classification within classes.

Specifically, we first extract the pixel-level spectral–spatial features from HSI. To reduce the size of the graph, we perform superpixel segmentation through simple linear iterative clustering (SLIC) and generate superpixel features [39]. Then, the kernel function [40] mapping is used to map the superpixel spectral–spatial features into a high-dimensional

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space (HDS), resulting in a more precise measurement of node connections. Subsequently, we employ a second-order biased random walk algorithm [41], which strikes a balance between DFS and BFS with bias, to construct a reliable hypergraph and characterize different community structure features. The hypergraph convolution and regular convolution are used for superpixel-level and pixel-level feature learning, respectively, to obtain classification results. Finally, to achieve balanced training of the two branches, the cross-entropy losses of CNN and HGCN are added as auxiliary components [39]. The main modules of CSGNet are described in detail below.

B. Feature Conversion

In this section, we extract the spectral–spatial features and achieve pixel-to-region and region-to-pixel assignments to fit the inputs of different feature extraction methods, including regular convolution and hypergraph convolution. Specifically, the original pixel spectral features of the HSI are represented as $\mathbf{X}_{\text{spe}} \in \mathbb{R}^{M\times L\times B}$, where M, L, and B denote the height, width, and the number of spectral channels [32]. The spatial features $\mathbf{X}_{\text{spa}} \in \mathbb{R}^{M\times L\times 2}$ are obtained through the pixel coordinates of the HSI, denoted as

$$\mathbf{X}_{\text{spa}}[i, j] = [i, j] \tag{1}$$

where i and j are the horizontal and vertical coordinates, respectively.

The pixel spectral–spatial features $\mathbf{X} = [\mathbf{X}_{\text{spe}}\ \mathbf{X}_{\text{spa}}]$ can be directly used as the inputs for CNN after data cleaning. In this article, we use two layers of 1×1 convolution for feature fusion across HSI channels and for denoising. For the HGCN, it is necessary to transform the HSI into a hypergraph. To reduce the size of the graph, we perform superpixel segmentation through SLIC and generate superpixel features. By introducing superpixels, multiple pixels are aggregated into a single unit. Even when individual pixels within the same class exhibit variability, the inductive nature of superpixels allows for the optimization of these variations. This approach helps mitigate the influence of noise and outliers, thereby improving classification accuracy within classes. In addition, inspired by Liu et al. [23], we apply HSI encode and decode operations to achieve pixel-to-region assignment and region-to-pixel assignment. Specifically, let $\mathbf{Q} \in \mathbb{R}^{ML\times N}$ be the transformation matrix between pixels and superpixels, where N denotes the number of superpixels, we have

$$\mathbf{Q}_{i,j} = \begin{cases} 1, & \text{if } \widetilde{\mathbf{X}}_i \in S_j \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

where $\widetilde{\mathbf{X}}$ represents the flattened spectral–spatial features $\mathbf{X}$ by the spatial dimension, $\widetilde{\mathbf{X}}_i$ is the ith pixel in $\widetilde{\mathbf{X}}$, and S_j is the jth superpixel. Subsequently, we achieve pixel-to-region assignment using the following formula:

$$\mathbf{V} = \text{Encode}(\mathbf{X}; \mathbf{Q}) = \widehat{\mathbf{Q}}^T\widetilde{\mathbf{X}} \tag{3}$$

where $\widehat{\mathbf{Q}}$ denotes the normalized transformation matrix by column. $\mathbf{V}$ is the superpixel features. Similarly, the region-to-pixel assignment can be achieved through the transformation matrix $\mathbf{Q}$, denoted as

$$\mathbf{X}^* = \text{Decode}(\mathbf{V}; \mathbf{Q}) = \mathbf{QV} \tag{4}$$

where $\mathbf{X}^*$ is the pixel features after region-to-pixel assignment.

C. Kernel Mapping and Global Relational Characterization

1) Kernel Mapping of Spectral Features: The kernel functions can map the original spectral–spatial features into a high-dimensional feature space, thereby alleviating the problem of spectral variability of land covers. Specifically, for the superpixel spectral–spatial feature $\mathbf{V} = \{\mathbf{v}_1, \cdots, \mathbf{v}_N\}$, we use the kernel function $k(\mathbf{v}_i, \mathbf{v}_j)$ to measure the similarity between any two superpixels $\mathbf{v}_i$ and $\mathbf{v}_j$. In particular, Gaussian radial basis functions (RBFs) can map data into an infinite dimensional space, making them well-suited for processing complex nonlinear spectral–spatial features, denoted as

$$\mathbf{A}_{i,j} = k(\mathbf{v}_i, \mathbf{v}_j) = \exp\left(-\frac{\| \mathbf{v}_i - \mathbf{v}_j \|^2}{2\sigma^2}\right) \tag{5}$$

where $\mathbf{A}_{i,j}$ denotes the similarity between nodes $\mathbf{v}_i$ and $\mathbf{v}_j$ in the high-dimensional space after kernel-mapped, and σ is a relaxation factor that controls the perceptual scale of the RBFs. A larger σ increases the receptive field of the kernel function, making the model more sensitive to global features of the data. Conversely, a smaller σ reduces the receptive field of the kernel function, making the model more sensitive to local features of the data. Therefore, we choose a combination of different σ values to help the RBF better capture the nonlinear features

$$\begin{aligned} \mathbf{A}_{i,j} &= \sum_{m=1}^{T} \beta_m k_m(\mathbf{v}_i, \mathbf{v}_j) \\ &= \sum_{m=1}^{T} \beta_m \exp\left(-\frac{\| \mathbf{v}_i - \mathbf{v}_j \|^2}{2\sigma_m^2}\right) \end{aligned} \tag{6}$$

where T denotes the number of RBF with different σ values and $\beta = \{\beta_1, \cdots \beta_T\}$ is the weight.

2) Global Relational and Scatter Structure Community Characterization: We use the global hypergraph $\mathcal{G}_{\text{global}}$ to characterize the global correlations in HSI. The global hypergraph $\mathcal{G}_{\text{global}}$ can be composed of a set of vertices $\mathcal{V}$ and a set of hyperedges $\mathcal{E}$. The correlations between vertex $\mathbf{v} \in \mathcal{V}$ and hyperedge $\mathbf{e} \in \mathcal{E}$ can be represented by an incidence matrix $\mathcal{H}$, denoted as

$$\mathcal{H}(\mathbf{v}, \mathbf{e}) = \begin{cases} 1, & \text{if } \mathbf{v} \in \mathbf{e} \\ 0, & \text{otherwise.} \end{cases} \tag{7}$$

Compared to the pairwise connections of simple graphs, hyperedges in hypergraphs can connect any number of nodes, thus characterizing complex high-order correlations. To construct the global hypergraph, for any given node $\mathbf{v}_i$, we find its k nearest-neighbors $\mathcal{N}_k(\mathbf{v}_i)$ according to the semantic similarity $\mathbf{A}_{i,j}$ and form a hyperedge $\mathbf{e}_i$, denoted as

$$e_i = \{\mathbf{v}_i, \forall \mathbf{v}_j \in \mathcal{N}_k(\mathbf{v}_i)\}. \tag{8}$$

Then, all hyperedges together form the global hypergraph $\mathcal{G}_{\text{global}}$. This process is applied to all nodes. Consequently, the scatter structure community that without specific obvious distribution characteristics are naturally represented.

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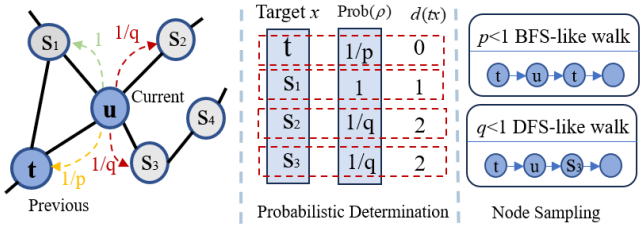


Fig. 3. Illustrations of hyperedge sampling process via second-order biased random walk, where t is the previous node, u is the current node, and x is the target node.

D. Community Structural Characterization

In this section, we characterize the spatial distribution semantic information of community structures contained in HSI, to enhance the reliability of the hypergraph and provide the model with richer semantic information. There are two main steps here: 1) community structure determination and 2) hyperedge node sampling. The detailed steps are as follows.

1) Community Structure Determination: Nodes with cluster structural features are usually connected to multiple nodes and have a higher degree. Therefore, based on [38], we design a cluster structure discriminator (CSD) using degree centrality. Specifically, for node $\mathbf{v}_i$ in $\mathbf{V} = \{\mathbf{v}_1, \cdots, \mathbf{v}_N\}$ and the matrix $\mathbf{A}_{ij}$ representing the similarity between nodes, its degree centrality $\mathbf{C}_d$ is defined as

$$\mathbf{C}_d(\mathbf{v}_i) = \sum_{j=1}^{N} \mathbf{A}_{ij} \tag{9}$$

then, the k_d nodes with the highest degree centrality are selected as the set of nodes with cluster structural features, denoted as $\mathcal{V}_d = \{\mathbf{v}_1, \cdots, \mathbf{v}_{k_d}\}$. Similarly, nodes with stripe structural features exhibit a linear distribution. Therefore, we designed a stripe structural discriminator (SSD) through betweenness centrality [38]. For node $\mathbf{v}_i$, its betweenness centrality $\mathbf{C}_b$ is typically denoted as

$$\mathbf{C}_b(\mathbf{v}_i) = \sum_{\mathbf{v}_s \neq \mathbf{v}_i \neq \mathbf{v}_t} \frac{\mathbf{O}_{st}(\mathbf{v}_i)}{\mathbf{O}_{st}} \tag{10}$$

where $\mathbf{O}_{st}$ represents the number of shortest paths from node s to node t, and $\mathbf{O}_{st}(\mathbf{v}_i)$ represents the number of those paths that pass through node i. Similarly, the k_b nodes with the highest betweenness centrality are selected as the set of nodes with stripe structural features, denoted as $\mathcal{V}_b = \{\mathbf{v}_1, \cdots, \mathbf{v}_{k_b}\}$.

2) Hyperedge Node Sampling: As discussed in Section I, what we need is a node sampling strategy that lies between DFS and BFS with some bias, to effectively characterize community structure distribution. Therefore, inspired by [41], we employ a second-order biased random walk algorithm, which strikes a balance between DFS and BFS, to comprehensively utilize information from different community structures. Specifically, as illustrated in Fig. 3, considering a sampling process that has just moved from node t to u, we need to determine the next sampled node x from node u. Define $\mathbf{w}_{u,x}$ as the edge weight and $\pi_{(u,x)}$ as the unnormalized transition probability

$$\pi_{u,x} = \rho_{t,x}\mathbf{w}_{u,x} \tag{11}$$

where

$$\rho_{t,x} = \begin{cases} \dfrac{1}{p}, & \text{if } d_{tx} = 0 \\ 1, & \text{if } d_{tx} = 1 \\ \dfrac{1}{q}, & \text{if } d_{tx} = 2 \end{cases} \tag{12}$$

p is the return parameter, which controls the likelihood of resampling the current node and the previous sampled nodes during the sampling process. q is in–out parameter, which controls the likelihood of sampling distant nodes. d_{tx} denotes the shortest path distance between nodes t and x, and the value must be one of $\{0, 1, 2\}$. Thus, the sampling process can be denoted as

$$P(c_i = x \mid c_{i-1} = u, c_{i-2} = t) = \begin{cases} \dfrac{\pi_{ux}}{Z}, & \text{if } \mathbf{A}_{u,x} \neq 0 \\ 0, & \text{otherwise} \end{cases} \tag{13}$$

where c_i denotes the ith node of the hyperedge sampling and Z is the normalizing constant.

Intuitively, the parameters p and q control the sampling direction (i.e., whether closer to or farther from the initial node u) and the sampling speed. The balance between DFS and BFS and the representation of community structures can be achieved by adjusting the parameters p and q.

BFS-Like Sampling: Parameter p controls the likelihood of resampling the current node and the previous sampled nodes. If p is set to a low value ($p < \min(q, 1)$), the sampling process tends to stay close to the current node, remaining within a local range. This strategy reflects the encouragement of local exploration by BFS, thereby effectively representing the cluster structure community. Similarly, if p is set to a higher value ($p > \max(q, 1)$), it reduces the likelihood of resampling nodes that have already been sampled.

DFS-Like Sampling: Parameter q controls the likelihood of sampling distant nodes. If q is set to a low value ($q < \min(p, 1)$), the process of hyperedge node sampling is more inclined to visit nodes that are further away from node t. Such sampling strategy is reflective of DFS, which encourages outward exploration, thereby effectively characterizing the stripe structure community. Similarly, if q is set to a higher value ($q > \max(p, 1)$), nodes nearest to t are more likely to be selected.

Parameter Selection and Community Structural Characterization: Flexibly adjusting the parameters p and q in the second-order biased random walk algorithm enables balancing between DFS and BFS for hypergraph construction. When $p = q = 1$, the biased random walk degenerates into a uniform random walk. Setting $p < \min(q, 1)$ encourages local exploration (i.e., BFS-like), while setting $q < \min(p, 1)$ promotes external exploration (i.e., DFS-like). Therefore, in this article, for the nodes with cluster structural features $\mathcal{V}_d = \{\mathbf{v}_1, \cdots, \mathbf{v}_{k_d}\}$, we set $p = 0.5$ and $q = 2$ to encourage local exploration to represent cluster structure community. Each sampled sequence $\chi_{\text{cluster}}(\mathbf{v}_i)$ from nodes $\mathbf{v}_i$ has similar cluster structural features and forms a hyperedge, denoted as $e_i = \{\mathbf{v}_i, \forall \mathbf{v}_i \in \chi_{\text{cluster}}(\mathbf{v}_i)\}$. Fusing the set of hyperedges $\mathcal{E}_{\text{cluster}} = \{\mathbf{e}_1, \mathbf{e}_2, \cdots, \mathbf{e}_{k_d}\}$ to obtain the cluster structure

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hypergraph $\mathcal{G}_{\text{cluster}} = (\mathcal{V}_{\text{cluster}}, \mathcal{E}_{\text{cluster}})$. Similarly, for the nodes with stripe structural features $\mathcal{V}_b = \{\mathbf{v}_1, \cdots, \mathbf{v}_{k_b}\}$, we set $p = 2$, $q = 0.5$ to encourage external exploration to represent stripe structure community. Each sampled sequence $\chi_{\text{stripe}}(\mathbf{v}_i)$ from nodes $\mathbf{v}_i$ has similar stripe structural features and forms a hyperedge, denoted as $e_i = \{\mathbf{v}_i, \forall \mathbf{v}_i \in \chi_{\text{stripe}}(\mathbf{v}_i)\}$. Fusing the set of hyperedges $\mathcal{E}_{\text{stripe}} = \{\mathbf{e}_1, \mathbf{e}_2, \cdots, \mathbf{e}_{k_b}\}$ to obtain the cluster structure hypergraph $\mathcal{G}_{\text{stripe}} = (\mathcal{V}_{\text{stripe}}, \mathcal{E}_{\text{stripe}})$. Finally, the cluster structure hypergraph $\mathcal{G}_{\text{cluster}}$ and the stripe structure hypergraph $\mathcal{G}_{\text{stripe}}$ are integrated into the global hypergraph $\mathcal{G}_{\text{global}}$, resulting in the final HSI hypergraph $\mathcal{G}$, represented by the association matrix as follows:

$$\mathcal{H} = \mathcal{H}_{\text{global}} \,||\, \mathcal{H}_{\text{cluster}} \,||\, \mathcal{H}_{\text{stripe}} \tag{14}$$

where $\mathcal{H}_{\text{global}}$, $\mathcal{H}_{\text{cluster}}$, and $\mathcal{H}_{\text{stripe}}$ represent the association matrices of the global hypergraph, the cluster structure hypergraph, and the stripe structure hypergraph, respectively.

E. Feature Learning and Classification

CSGNet is a dual-branch structure. First, CSGNet utilizes hypergraph convolution [26] to learn high-order features. The process of hypergraph convolution can be divided into two parts, hyperedge convolution and vertex convolution, denoted as

$$\mathbf{Y} = \mathbf{D}_v^{-\frac{1}{2}} \mathcal{H} \mathbf{W} \mathbf{D}_e^{-1} \mathcal{H}^T \mathbf{D}_v^{-\frac{1}{2}} \mathbf{V}^{(l)} \Theta^{(l)} \tag{15}$$

where $\mathbf{D}_v$ and $\mathbf{D}_e$ represent the diagonal matrices of vertex degrees and edge degrees, $\mathbf{W}$ denotes the weight matrix of the hyperedges, and Θ is the parameter to be learned. Hyperedge convolution is achieved by multiplying $\mathbf{V}$ by $\mathcal{H}$, aggregating vertex features into the hyperedges. Then, by multiplying with $\mathcal{H}^T$, the hyperedge features are propagated back to the nodes, updating their features accordingly. In this article, the number of hypergraph convolution layers is set to 2, and the final output is denoted as $\mathbf{Y}_{\text{HGCN}}$. Meanwhile, the CNN branch, which includes two layers of spectral–spatial convolution (SSConv) [23], is introduced for pixel-level local information extraction. By adding this branch, the loss of pixel-level spectral–spatial features can be reduced. The output of the CNN is denoted as $\mathbf{Y}_{\text{CNN}}$. The detailed parameters of HGCN and CNN branches are described in experimental setting. Through (4), we achieve the feature transformation from superpixels to pixels, unifying the results of hypergraph convolution and regular convolution into a single dimension.

During the fusion of the two branches, we assign different weights to better integrate the model

$$\mathbf{Y} = \eta \mathbf{Y}_{\text{CNN}} + (1 - \eta) \mathbf{Y}_{\text{HGCN}} \tag{16}$$

where η is the weight of CNN. We use the cross-entropy loss function to train CSGNet, denoted as

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{C} \mathbf{L}_j(p_i) \log\left(\mathbf{Y}_j(p_i)\right) \tag{17}$$

where N is the number of samples, C is the number of classes, $\mathbf{L}_j(p_i)$ is the jth element of the label $\mathbf{L}(p_i)$, and $\mathbf{Y}_j(p_i)$is the probability that the pixel belongs to the jth class.

TABLE I
CATEGORY INFORMATION OF THE PAVIAU

Class id	Color	Class name	Training	Validation	Testing
1		Asphalt	7	7	6617
2		Meadows	19	19	18611
3		Gravel	3	3	2093
4		Trees	4	4	3056
5		Metal Sheets	2	2	1341
6		Bare Soil	6	6	5017
7		Bitumen	2	2	1326
8		Bricks	4	4	3674
9		Shadows	1	1	945
Total			48	48	42680

To achieve balanced training of the HGCN and CNN branches, we add the cross entropy of $\mathcal{L}_{\text{CNN}}$ and $\mathcal{L}_{\text{HGCN}}$ as an auxiliary component to participate in the backpropagation, denoted as

$$\mathcal{L}\text{oss} = \mathcal{L}_{\text{Fusion}} + \mathcal{L}_{\text{CNN}} + \mathcal{L}_{\text{HGCN}} \tag{18}$$

where $\mathcal{L}_{\text{Fusion}}$ is the cross-entropy loss of $\mathbf{Y}$. $\mathcal{L}_{\text{CNN}}$ and $\mathcal{L}_{\text{HGCN}}$ is the cross-entropy loss of CNN and HGCN branches, respectively.

III. EXPERIMENTAL

To validate the performance of CSGFNet, three benchmark HSI datasets are used for comparative experiments, namely, the University of Pavia (PaviaU), Xuzhou, and the WHU-Hi-HongHu (WHU) datasets. Moreover, nine deep learning methods are chosen for comparison, including multiscale 3-D deep CNN (M3DCNN) [10], HybridSN [12], SSRN [14], RNN [42], GCN [18], F2HGNNss [33], CEGCN [23], AMGCFN [39], and MambaHSI [43]. Among these methods, M3DCNN, HybridSN, and SSRN are CNN-based networks. GCN and F2HGNNss are graph-based and hypergraph-based networks, respectively. CEGCN and AMGCFN are hybrid networks, integrating CNN and GCN into an end-to-end framework. The per-class accuracy, overall accuracy (OA), average accuracy (AA), and kappa coefficient (KC) are used to quantitatively evaluate the performance of the network. All reported experimental results are the mean values of ten independent runs.

A. Dataset Description

1) PaviaU Dataset: This HSI dataset [44] was collected by the ROSIS sensor of the PaviaU in northern Italy. This HSI dataset consists of 640 × 340 pixels and 103 spectral bands, with data labeled into nine distinct land cover classes for analysis and experimentation. This dataset exhibits distinct spatial distribution characteristics of land covers, including numerous typical stripe structure communities and cluster structure communities. In this data, 0.1%, 0.1%, and 99.8% of samples from each class are randomly selected for training, validation, and testing, respectively. The detailed category information is shown in Table I.

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TABLE II
CATEGORY INFORMATION OF XUZHOU DATASET

Class id	Color	Class name	Training	Validation	Testing
1		Bareland1	27	27	26342
2		Lakes	5	5	4017
3		Coals	3	3	2777
4		Cement	6	6	5202
5		Crops-1	14	14	13156
6		Trees	3	3	2430
7		Bareland2	7	7	6976
8		Crops	5	5	4767
9		Red-title	4	4	3062
	Total		74	74	68729

TABLE III
CATEGORY INFORMATION OF THE WHU DATASET

Class id	Color	Class name	Training	Validation	Testing
1		Chinese cabbage	4	4	3312
2		Garlic sprout	2	2	1605
3		Rape	22	22	21777
4		Road	30	30	29234
5		Carrot	42	42	41439
6		White radish	18	18	17826
7		Lactuca sativa	5	5	4044
8		Tuber mustard	11	11	10284
9		Bare soil	13	13	12368
10		Cotton	7	7	6533
11		Film covered lettuce	9	9	8569
12		Cabbage	19	19	18882
13		Brassica parachinensis	8	8	7340
14		Pakchoi	2	2	998
15		Romaine lettuce	8	8	7246
16		Red roof	4	4	3002
17		Celtuce	4	4	3209
18		Brassica chinensis	5	5	4191
19		Small brassica chinensis	4	4	3478
	Total		217	217	205337

2) Xuzhou Dataset: This HSI dataset [45] was collected by an airborne HYSPEX hyperspectral camera over Xuzhou. The HSI dataset contains 436 bands, and the size is 500 × 260, and labeled into nine distinct land cover classes. In this data, 0.1%, 0.1%, and 99.8% of labeled samples per class are chosen as the training, validation, and testing sets, respectively. The detailed category information is shown in Table II.

3) WHU Dataset: This HSI dataset [36] was collected by a Headwall Nano-Hyperspectral imaging sensor equipped on a DJI Matrice 600 Pro UAV platform over HongHu city, China, in 2017. We intercepted the region with a size of 490 × 475 pixel and there are 270 bands from 400 to 1000 nm, labeled into 19 different classes, are used for analysis and experimentation. The agricultural landscape of the study area is complex, with a wide variety of crops and different cultivars of the same species, including Chinese cabbage and cabbage, as well as brassica chinensis and small brassica chinensis also being cultivated in the region, exhibiting significant spectral heterogeneity. 0.1%, 0.1%, and 99.8% of labeled samples per class are chosen as the training, validation, and testing sets, respectively. The detailed category information is shown in Table III.

B. Experimental Setting

The superpixel segmentation scale λ in feature conversion is set to 100, meaning that each superpixel region contains approximately 100 pixels. The number of key nodes k_d and k_b is set to 10% of the total number of superpixels. The fusion weight η is set to 0.6 for the PaviaU dataset, 0.8 for Xuzhou dataset, and 0.2 for the WHU dataset. The number of σ is set to 5, and be valued at 0.1, 0.5, 1, 1.5, 2. The corresponding weights β are set to 0.1, 0.35, 0.35, 0.1, 0.1. The architecture configuration of the CNN and HGCN are summarized in Table IV. All networks in this article are implemented in the PyTorch framework with an initial learning rate of 0.001. We adopt the Adam optimizer with weight decay of 0.0005 to train network and the number of training iterations is set to 500.

TABLE IV
ARCHITECTURE CONFIGURATION OF THE CNN AND HGCN

Module	Layer no.	Kernel Size
CNN	1	5×5@128
	2	3×3@64
HGCN	1	@128
	2	@64

C. Comparison of Classification Performance

In this section, we analyze the classification performance of all networks on three HSI benchmark datasets. The quantitative classification results of different networks are displayed in Tables V–VII, in which the highest values are in bold. Correspondingly, the classification maps obtained by these methods are shown in Figs. 4–6.

1) PaviaU Dataset: In this data, first, we can observe that CNN-based methods have a strong dependency on the training samples. For example, M3DCNN and HybridSN achieved OA of 80.36% and 76.84%, respectively. Additionally, the issue of limited training samples also has a greater impact on the latest MambaHSI, which achieves an OA of only 87.29%. In contrast, graph-based models exhibit more stable performance when the number of labeled samples is limited. For example, GCN achieve OA of 86.44% on the PaviaU dataset, which is 6% higher than M3DCNN. Compared to simple graphs, F2HGNN utilizing hypergraphs further improves classification performance, achieving an OA of 89.20%. This improvement is primarily attributed to the superior representational capacity of hypergraphs over simple graphs. While simple graphs tend to focus on pairwise relationships between nodes, hypergraphs can model relationships among any number of nodes, allowing for more effective representation of high-order nonlinear spectral–spatial features. It is worth noting that the proposed CSGNet achieves outstanding performance with an OA of 94.64%, an AA of 92.16%, and a KC of 92.86%. This can primarily be attributed to two factors. First, CSGNet leverages the hypergraph to utilize unlabeled samples. The propagation mechanism of hypergraph convolution transfers node features to adjacent superpixels, allowing unlabeled pixels to also benefit from the information provided by labeled samples. On the other hand, CSGNet improves edge detection. It models the spatial distribution structure of features through hypergraphs (i.e., nodes with similar distribution structures are connected

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TABLE V

CLASSIFICATION RESULTS (%) ON THE PAVIAU DATASET WITH 0.1% TRAINING SAMPLES PER CLASS. THE HIGHEST ACCURACIES ARE HIGHLIGHTED IN BOLD

Class ID	M3DCNN [10]	HybridSN [12]	SSRN [14]	RNN [42]	GCN [18]	F2HGNNss [33]	CEGCN [23]	AMGCFN [39]	MambaHSI [43]	CSGNet (ours)
1	83.88±1.31	49.87±7.65	85.71±4.08	74.18±3.71	81.66±5.55	88.03±5.08	89.26±10.35	**90.21±4.47**	83.14±7.15	87.35±3.81
2	85.48±6.34	93.23±4.24	91.88±3.28	81.65±6.98	93.69±3.40	97.29±2.39	96.74±1.96	97.90±1.32	97.00±1.89	**99.67±0.39**
3	42.15±13.34	44.26±5.29	37.75±15.06	27.36±11.52	69.48±5.12	64.79±26.09	86.95±6.80	74.15±15.60	55.70±10.05	**90.78±12.31**
4	85.91±7.32	84.95±2.05	**95.14±1.97**	81.54±9.22	65.01±10.32	80.29±6.57	75.12±11.72	68.01±8.86	76.42±8.38	78.92±13.59
5	93.35±7.11	99.38±1.24	99.32±0.75	94.54±8.88	98.64±1.65	99.18±1.44	**100.00±0.00**	99.55±0.92	96.62±4.73	**100.00±0.00**
6	68.61±3.02	94.34±2.13	92.29±5.94	49.67±18.31	95.23±3.84	90.43±8.26	93.38±10.22	94.77±4.71	83.26±12.50	**99.63±0.52**
7	76.36±2.89	30.50±11.33	84.97±14.55	22.11±9.40	73.35±23.12	84.33±11.04	85.42±22.55	73.38±19.46	57.84±25.95	**98.24±0.69**
8	76.81±9.45	54.83±18.13	89.13±11.37	78.75±11.88	84.60±8.04	72.26±19.89	78.14±20.01	81.51±7.75	82.45±7.43	**89.94±0.47**
9	84.75±15.01	14.74±10.48	**99.85±0.12**	75.84±22.31	44.42±10.29	72.83±2.88	58.88±23.74	50.85±18.93	98.76±0.89	84.94±3.84
OA	80.36±3.39	76.84±1.43	88.54±1.62	72.23±3.05	86.44±1.82	89.20±1.81	90.47±3.50	89.87±2.25	87.29±1.43	**94.64±2.48**
AA	77.48±2.87	62.90±3.08	86.34±2.22	65.07±1.70	78.48±3.54	83.27±2.88	84.88±6.52	81.15±4.42	81.25±2.62	**92.16±3.70**
KC	74.20±4.22	69.54±1.77	84.98±2.02	63.47±3.60	82.14±2.29	85.65±2.39	87.33±4.75	86.46±3.05	79.36±5.29	**92.86±3.33**

TABLE VI

CLASSIFICATION RESULTS (%) ON XUZHOU DATASET WITH 0.1% TRAINING SAMPLES PER CLASS. THE HIGHEST ACCURACIES ARE HIGHLIGHTED IN BOLD

Class ID	M3DCNN [10]	HybridSN [12]	SSRN [14]	RNN [42]	GCN [18]	F2HGNNss [33]	CEGCN [23]	AMGCFN [39]	MambaHSI [43]	CSGNet (ours)
1	95.13±1.26	**96.36±0.26**	92.28±1.66	92.41±0.45	94.14±1.87	95.53±1.97	94.81±1.21	94.80±1.52	95.11±0.37	96.16±0.74
2	96.25±0.58	89.30±2.45	96.86±2.82	96.65±1.16	95.32±2.33	97.97±2.17	99.72±0.34	98.45±0.89	98.16±0.26	**99.89±0.10**
3	71.02±6.47	94.70±1.70	72.79±25.86	74.09±9.27	94.13±8.50	92.52±5.63	93.01±10.22	93.78±9.55	87.58±9.88	**99.75±0.24**
4	82.38±6.92	86.26±5.56	87.89±11.32	84.35±2.79	68.57±3.39	78.45±7.73	86.16±6.42	83.48±13.18	81.39±11.1	**96.94±3.37**
5	93.68±2.32	95.14±2.42	93.60±4.36	84.70±2.67	95.85±1.29	96.99±1.77	97.79±1.18	97.16±2.56	96.84±1.04	**99.32±0.16**
6	18.05±7.39	31.35±4.46	58.95±33.84	10.27±2.66	91.33±6.80	67.66±22.60	74.86±14.12	87.15±10.95	78.09±12.82	**92.91±2.87**
7	72.17±6.83	92.27±2.58	89.69±6.79	55.88±3.39	94.58±2.24	94.19±3.69	94.97±3.40	97.95±1.36	97.20±2.85	**98.57±0.49**
8	93.37±5.51	98.32±0.69	93.66±13.31	95.82±2.74	93.70±5.39	95.76±2.16	**100.00±0.00**	99.32±0.64	98.43±1.05	99.86±0.06
9	87.90±4.95	79.47±14.07	**94.95±2.77**	89.21±6.21	72.72±13.41	79.70±5.47	93.02±3.10	86.69±8.26	89.56±3.39	94.62±1.91
OA	87.48±1.33	91.93±1.39	90.45±2.03	83.31±1.15	91.56±0.38	92.83±1.30	94.52±0.64	94.57±0.99	93.87±0.62	**97.50±0.54**
AA	78.88±2.16	84.91±2.49	86.74±4.38	75.93±2.06	88.93±1.01	88.87±3.24	92.70±1.09	93.20±1.87	91.37±1.85	**97.56±0.67**
KC	83.96±1.72	89.72±1.76	87.98±2.55	78.73±1.46	89.31±0.45	90.90±1.65	93.06±0.82	93.13±1.24	90.79±4.04	**96.84±0.68**

TABLE VII

CLASSIFICATION RESULTS (%) ON THE WHU DATASET WITH 0.1% TRAINING SAMPLES PER CLASS. THE HIGHEST ACCURACIES ARE HIGHLIGHTED IN BOLD

Class ID	M3DCNN [10]	HybridSN [12]	SSRN [14]	RNN [42]	GCN [18]	F2HGNNss [33]	CEGCN [23]	AMGCFN [39]	MambaHSI [43]	CSGNet (ours)
1	67.51±21.95	94.21±3.99	84.42±9.03	23.72±10.08	86.38±3.07	92.38±0.58	**96.27±1.75**	92.01±4.34	84.85±8.52	94.92±1.43
2	77.35±2.38	74.19±11.53	**78.31±4.04**	53.16±0.34	32.44±8.45	44.67±2.37	59.00±17.51	74.66±11.39	50.9±17.04	20.27±3.87
3	91.75±1.03	89.62±1.06	94.51±0.37	90.95±2.39	86.28±1.29	93.58±0.33	**97.71±0.52**	94.02±2.20	96.09±0.49	97.30±0.33
4	99.00±0.16	99.62±0.49	99.01±0.61	94.57±0.59	99.27±0.31	99.73±0.04	**99.91±0.01**	99.28±0.60	99.58±0.16	98.56±1.11
5	96.56±1.53	97.86±0.77	97.59±0.41	90.27±0.98	98.04±0.70	98.66±0.37	99.48±0.08	**99.53±0.29**	98.51±0.87	98.15±0.70
6	70.58±8.38	86.82±0.53	82.75±7.73	65.86±0.97	92.98±1.47	93.90±0.56	93.22±2.67	94.36±0.81	87.71±1.21	**94.69±0.23**
7	17.59±7.00	47.90±5.62	23.45±11.11	7.19±1.15	67.80±10.34	80.02±1.94	34.14±25.06	79.22±5.08	40.1±17.76	**89.10±3.38**
8	89.86±4.50	99.06±0.26	98.11±10.00	86.20±2.00	93.41±0.37	95.90±0.86	**98.71±0.67**	97.19±3.83	88.87±4.91	98.17±0.11
9	67.59±5.16	86.95±4.47	76.61±1.28	23.14±6.43	**92.77±1.69**	85.27±1.95	91.02±1.38	88.62±4.71	76.5±2.37	89.47±0.87
10	52.10±9.21	86.85±4.65	70.13±8.30	29.34±1.07	79.96±4.21	79.67±3.00	80.15±4.28	**88.72±6.19**	74.25±2.30	83.96±1.03
11	58.38±2.55	66.82±2.09	50.10±2.70	39.82±3.06	69.49±4.53	75.33±1.99	81.10±5.74	84.63±5.99	69.86±3.88	**85.82±1.33**
12	72.08±4.71	83.24±2.16	78.38±2.53	58.45±3.39	84.51±3.86	90.70±1.06	91.69±0.08	89.31±2.42	90.27±5.55	**96.98±0.17**
13	63.42±11.54	**92.51±1.87**	80.36±5.59	42.27±1.73	89.00±0.97	90.65±3.86	70.87±5.62	85.67±10.06	65.79±10.56	86.46±1.15
14	60.88±8.04	88.15±0.67	77.61±8.00	14.04±14.51	75.45±5.65	36.07±12.59	78.99±6.15	80.37±15.31	73.31±6.76	**94.73±1.12**
15	92.97±4.15	92.40±0.38	96.10±2.40	60.68±8.33	83.91±4.53	87.03±2.47	94.67±1.31	91.26±5.92	88.25±7.39	**98.12±0.53**
16	31.95±22.2	70.21±1.47	55.92±2.61	51.79±10.05	77.43±1.58	79.09±1.06	**83.81±8.92**	74.85±10.40	60.47±14.77	48.29±3.20
17	55.48±9.39	81.90±9.13	61.57±7.95	25.32±13.85	54.08±6.46	84.79±3.74	66.76±9.36	88.27±2.88	64.08±16.40	**94.23±0.27**
18	42.61±15.38	67.34±10.73	48.81±6.68	13.38±3.10	70.67±1.31	73.82±0.51	**75.52±14.02**	73.00±16.66	68.83±5.99	65.07±3.57
19	39.62±4.83	61.16±5.01	45.27±20.54	15.18±10.8	73.57±2.07	84.43±3.37	73.92±16.32	76.73±16.02	56.09±4.02	**95.60±1.81**
OA	79.49±3.53	89.13±0.13	84.67±0.79	67.07±1.62	88.78±0.81	91.46±0.10	91.55±0.97	92.71±0.81	87.12±0.47	**93.44±0.22**
AA	65.65±6.08	82.46±1.08	73.63±1.99	46.60±3.20	79.33±0.89	82.40±0.80	82.47±2.49	85.63±1.85	75.49±1.65	**85.78±0.31**
KC	77.09±3.94	87.91±0.15	82.87±0.89	62.97±1.84	87.50±0.90	90.48±0.10	90.56±1.09	91.88±0.91	81.32±1.67	**92.68±0.25**

by a single hyperedge), providing richer contextual information for the model and thus enhancing its edge detection capability.

2) Xuzhou Dataset: For Xuzhou dataset, which has a higher sample labeling rate, the OA values of the models are generally higher than those of the PaviaU dataset. For instance, M3DCNN, HybridSN, and SSRN achieve OAs of 87.48%, 91.93%, and 90.45%, respectively. However, from Fig. 5(c) and (e), we can observe that the classification results of M3DCNN and SSRN exhibit significant salt-and-pepper noise, while graph-based methods such as GCN exhibit much smoother results. It is worth noting that hybrid models can effectively utilize the advantages of different networks to improve classification robustness. Graph-based methods

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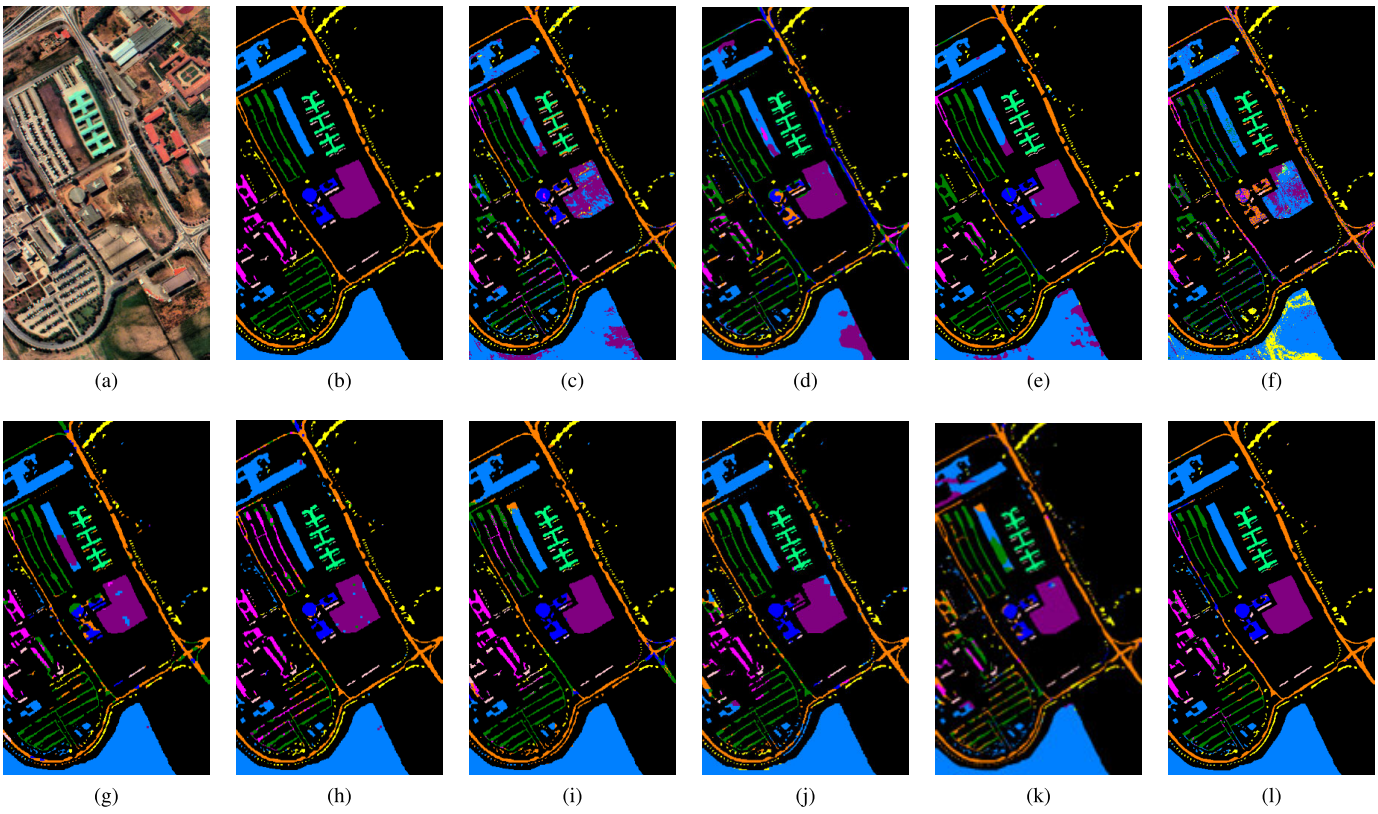


Fig. 4. Classification maps of different methods for the PaviaU dataset. (a) False-color image. (b) Ground truth. (c) M3DCNN. (d) HybirdSN. (e) SSRN. (f) RNN. (g) GCN. (h) F2HGNNss. (i) CEGCN. (j) AMGCFN. (k) MambaHSI. (l) CSGNet (ours).

largely rely on superpixel nodes, and during the superpixel segmentation process, a significant amount of pixel-level features is inevitably lost. As shown in Fig. 5(g) and (h), GCN and F2HGNN perform poorly in classifying small areas. In contrast, hybrid networks, such as CEGCN and AMGCFN, integrate CNN and GCN within an end-to-end deep learning framework, achieving OA of 94.52% and 94.57%, effectively reducing pixel-level features loss. The proposed CSGNet achieved an OA of 97.50%, further improving by 2% compared to CEGCN and AMGCFN. This is also attributed to the stronger representational capability of hypergraphs compared to simple graphs. Additionally, the modeling of spatial distribution semantic information of land covers provides CSGNet with richer contextual semantic information.

3) WHU Dataset: This data contain a diverse range of crop types, including multiple species and different varieties of the same species, such as Chinese cabbage and cabbage, as well as brassica chinensis and small brassica chinensis, exhibiting significant spectral heterogeneity. From Table VII and Fig. 6, it can be observed that methods relying on spectral features, such as M3DCNN, achieves only 79.49%, which can struggle to distinguish between adjacent classes with similar spectral features, especially in heterogeneous regions where edges are subtle or blurred. In contrast, F2HGNNss constructs spatial hypergraphs and jointly utilizes spectral–spatial features, achieving an OA of 91.46%, which is 3% higher than GCN. AMGCFN leverages multiscale graph and attention mechanisms to enhance the learning of contextual semantic information, achieving an OA of 92.71%, which is a 1% improvement over CEGCN. CSGNet achieves the highest OA of 93.44%. Without incorporating attention mechanisms or multiscale graph structures, it outperforms AMGCFN in classification performance. This phenomenon also suggests that CSGNet, by modeling the semantic information of community structure spatial distribution, provides richer contextual information for the model, thereby improving its ability for edge detection. At the same time, CSGNet employs the kernel function to map the original spectral–spatial features of superpixels into a high-dimensional feature space, further improving classification within classes.

IV. Discussion

A. Ablation Analysis

The proposed CSGNet is a dual-branch structure, comprising a CNN branch and an HGCN branch. Meanwhile, the HGCN branch includes two essential modules: a kernel function mapping module for nonlinear spectral features mapping, and a random walk module for community structure representation. To evaluate the performance of each module, we conducted ablation experiments on the PaviaU, XuZhou, and WHU datasets. Specifically, we first compared the performance of individual branches [i.e., experiments (1) and (2)]. Then, we evaluated the performance after fusing the CNN and HGCN branches [i.e., experiment (3)]. Subsequently, we added the kernel mapping module [experiment (4)] and the random walk module [experiment (5)] separately. Finally, we fused all modules [experiment (6)]. The detailed experimental results are presented in Table VIII.

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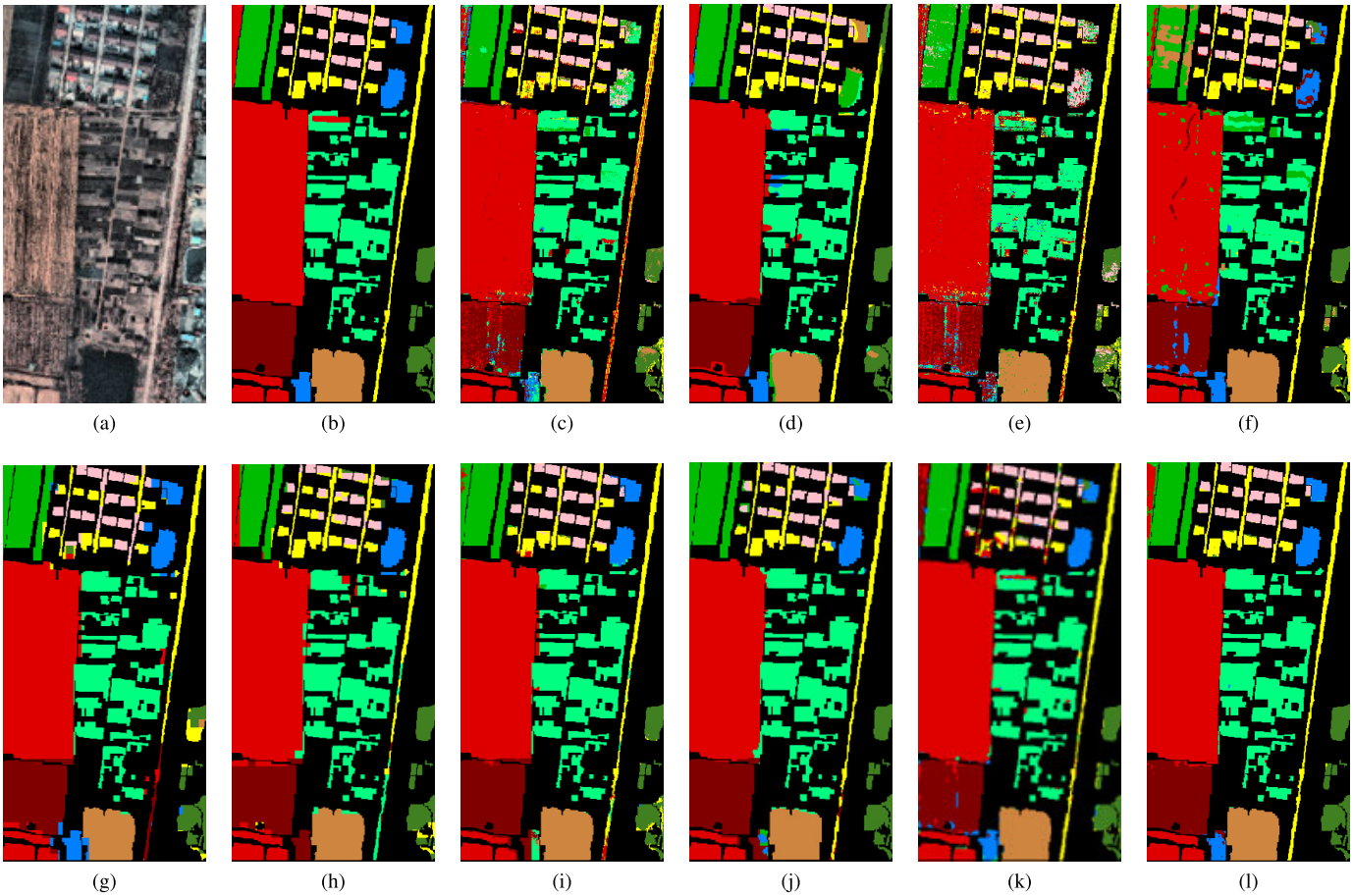


Fig. 5. Classification maps of different methods for Xuzhou dataset. (a) False-color image. (b) Ground truth. (c) M3DCNN. (d) HybirdSN. (e) SSRN. (f) RNN. (g) GCN. (h) F2HGNNss. (i) CEGCN. (j) AMGCFN. (k) MambaHSI. (l) CSGNet (ours).

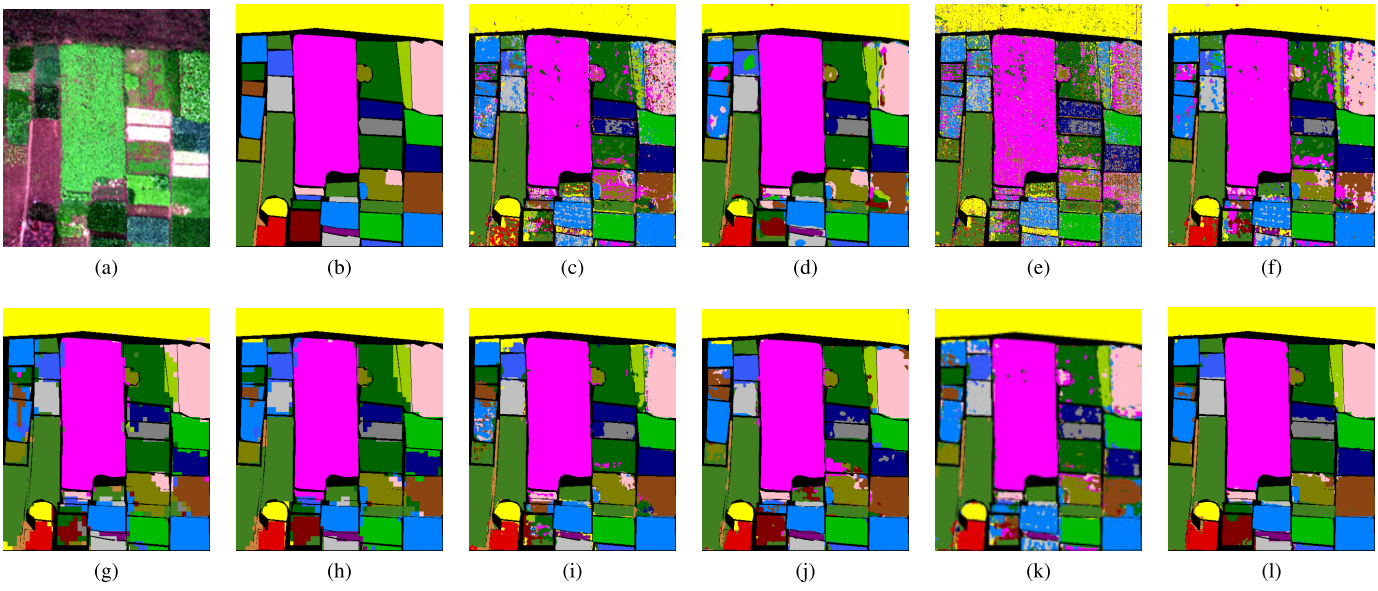


Fig. 6. Classification maps of different methods for the WHU dataset. (a) False-color image. (b) Ground truth. (c) M3DCNN. (d) HybirdSN. (e) SSRN. (f) RNN. (g) GCN. (h) F2HGNNss. (i) CEGCN. (j) AMGCFN. (k) MambaHSI. (l) CSGNet (ours).

We can observe that (1) achieves higher OA than (2) on the PaviaU dataset, while (1) shows lower classification performance compared to (2) on the WHU dataset. In Xuzhou dataset, both branches exhibit similar accuracy. This difference is due to the complex distribution of ground objects in the PaviaU dataset, where regions with the same label are relatively small. As a result, the pixel purity within superpixel blocks is low, and the similarity between nodes varies significantly, thereby affecting the feature extraction capabilities of HGCN. From the results of (3), we can see that the CNN

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TABLE VIII
ABLATION STUDY RESULTS (%) ON PAVIAU, XUZHOU, AND WHU DATASETS

Model	(1)	(2)	(3)	(4)	(5)	(6)
CNN	✓		✓	✓	✓	✓
HGCN		✓	✓	✓	✓	✓
Kernel mapping				✓		✓
Rand walk					✓	✓
PaviaU	90.03±1.07	86.95±0.15	92.85±0.90	93.70±0.20	93.26±0.69	94.64±2.48
Xuzhou	93.21±0.30	93.89±0.07	95.98±0.19	97.07±0.19	96.07±0.29	97.50±0.54
WHU	86.95±0.74	92.39±0.14	92.57±0.12	93.07±0.27	92.91±0.14	93.44±0.22

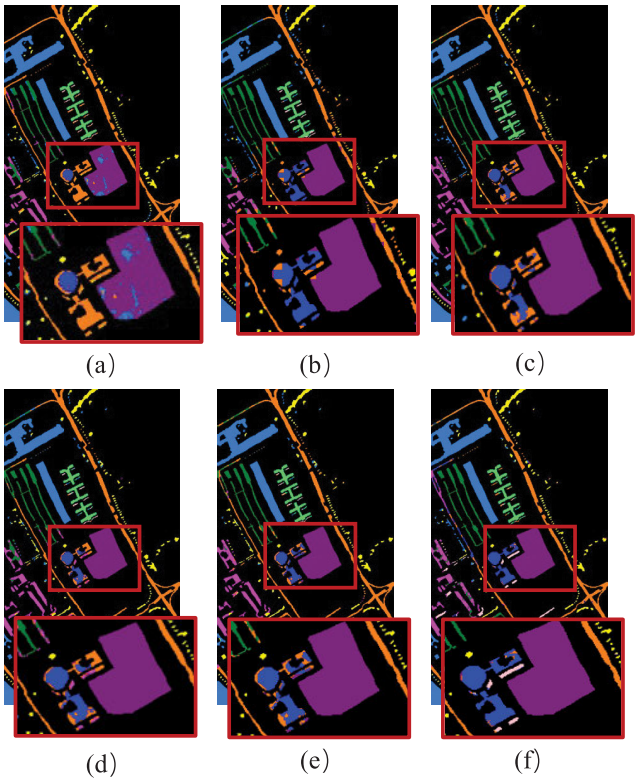


Fig. 7. Classification maps of different modules for the PaviaU dataset. (a) Experiment (1). (b) Experiment (2). (c) Experiment (3). (d) Experiment (4). (e) Experiment (5). (f) Experiment (6).

can complement the HGCN with pixel-level features, thus improving the classification performance. (4) achieves more accurate node connections and alleviates the spectral nonlinear redundancy problem through kernel functions. (5) represents the spatial distribution semantic information of community structures by second-order biased random walk. This enhances the reliability of the hypergraph in HSI and improves the learning ability of HGCN for high-order correlations. Finally, (6) achieves the best classification performance by integrating all the modules. We selected a typical region from the PaviaU dataset to compare the classification performance of different modules, as shown in Fig. 7. It can be observed that, with the enhancement of the kernel mapping and community structure features, the proposed CSGNet achieves more accurate classification results.

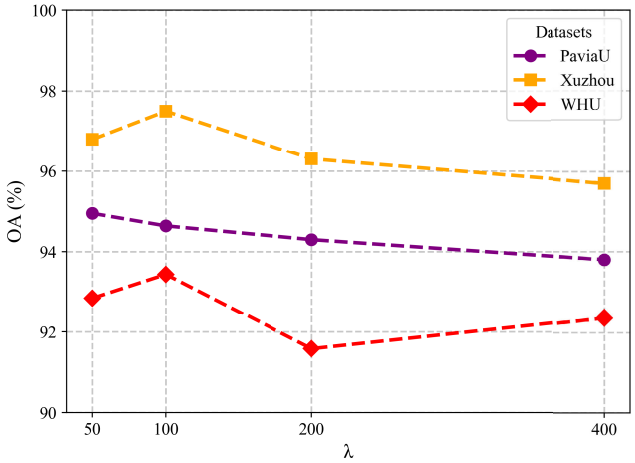


Fig. 8. OAs under different segmentation scale λ on each dataset.

B. Influence of Segmentation Scale λ

The segmentation scale λ is directly associated with the relationship between pixels and superpixels. As λ increases, the number of pixels corresponding to a single superpixel region grows, implying a greater loss of pixel-level features. Therefore, we evaluate the classification performance of the model with λ set to 50, 100, 200, and 400 on three dataset. The experimental results are shown in Fig. 8. It can be observed that by balancing the training of the CNN and HGCN networks, CSGNet demonstrates stable performance with only a slight decrease in OA as λ increases.

C. Influence of Training Sample Ratio

Because pixel labeling of HSI is expensive, we used 0.1% of the samples for training in previous experiments. The results demonstrate that the proposed CSGNet performs well when the training sample is limited. In this section, to validate the robustness of the model, we tested the performance of CSGNet and comparison methods on three datasets with different training sample ratios. The training ratios were set to 0.1%, 0.2%, 0.3%, 0.4%, and 0.5%. The experimental results are

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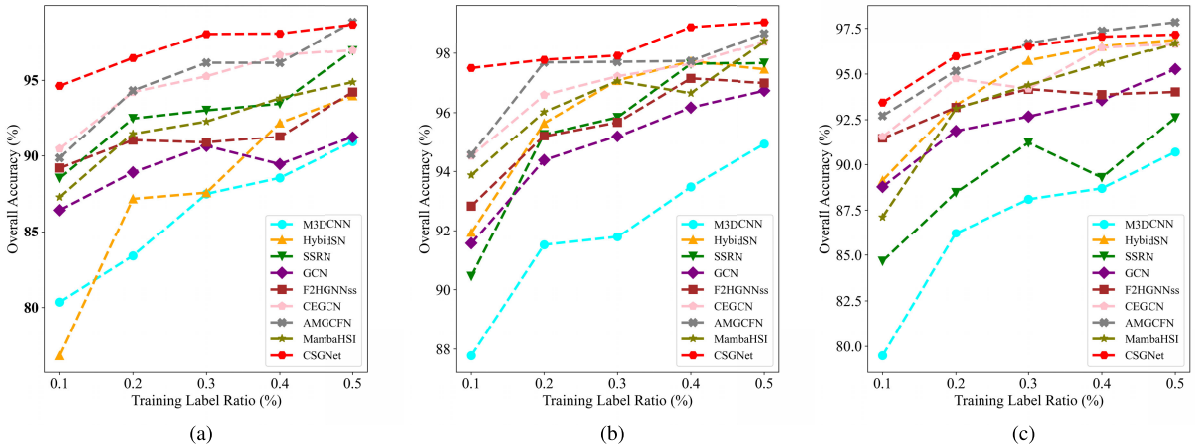


Fig. 9. Classification accuracy with different sample ratios. (a) PaviaU dataset. (b) Xuzhou dataset. (c) WHU dataset.

shown in Fig. 9. From the above experimental results, we can observe that the OAs of all the methods are improved with the increase in the training samples. For example, the OA of GCN improves from 88.78% to 95.10% as the training ratio increases from 0.1% to 0.5% on WHU dataset. Additionally, several patterns can also be observed. First, F2HGNNss consistently outperforms GCNs, demonstrating that hypergraphs are more powerful representations than simple graphs. Second, the performance of hybrid networks has better results than single-architecture networks at different training ratios. In particular, AMGCFN achieves the best classification performance on the WHU dataset with a training ratio of 0.5%. This is attributed to the ability to mine spectral–spatial features at different scales through multiscale GCNs and efficiently fuse them together through an attention fusion module. The proposed CSGNet achieves better classification performance with different training ratios. Particularly, the performance improvement is more obvious when the training samples are limited. This is attributed to two factors: first, CSGNet leverages the hypergraph to exploit potential information from unlabeled samples. Second, CSGNet represents the spatial distribution information of community structures, providing richer contextual information to the model, which enhances the reliability of the hypergraph and improves both the ability of edge detection and classification within classes.

TABLE IX
PARAMETER CHOICES SCHEMES FOR SECOND-ORDER BIASED RANDOM WALK SETTING

Schemes	Cluster structure		Stripe structure	
	p	q	p	q
Schemes-1	1	1	1	1
Schemes-2	0.5	2	2	0.5
Schemes-3	2	0.5	0.5	2

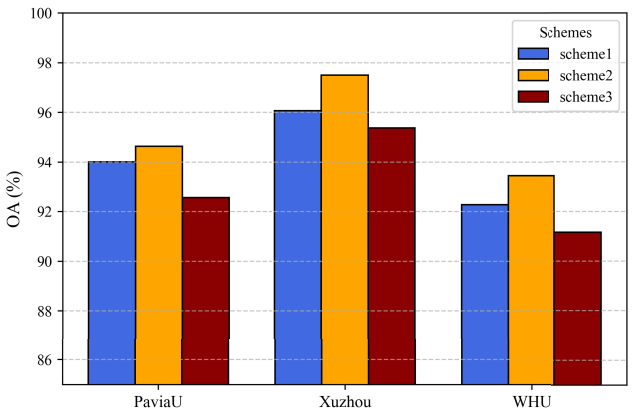


Fig. 10. OAs under different schemes on each dataset.

D. Influence of Second-Order Biased Random Walk

To verify the correctness of the parameter choices for p and q of second-order biased random walk, three schemes are designed, as shown in Table IX. Scheme-1 is defined as $p = q = 1$, which means the biased random walk degenerates into a uniform random walk. Scheme-2 is the strategy adopted by CSGNet. For the nodes with cluster structural features, $p = 0.5$ and $q = 2$, while for the nodes with stripe structural features, $p = 2$ and $q = 0.5$. Scheme-3 adopts the opposite strategy to Scheme-2. The experimental results are presented in Fig. 10. It is observed that Scheme-2 achieves the highest OA, while Scheme-3 obtains the lowest OA. The results demonstrate that when $p < \min(q, 1)$, the sampling process tends to favor DFS, while when $q < \min(p, 1)$, it tends to favor BFS.

E. Comparison of Computational Efficiency

In this section, we evaluate the operational efficiency and computational cost of the model. Hypergraph construction is a crucial step in the classification process. Employing second-order biased random walks to construct hypergraphs effectively captures intrinsic semantic information while minimizing resource consumption. Specifically, second-order biased random walks require storing second-order association information between neighbors, resulting in a space complexity of $O(a^2V)$, where $O(a^2)$ denotes the storage requirement

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TABLE X
TRAINING TIME (S) AND PARAMETERS (K) COMPARISON ON PAVIAU, XUZHOU, AND WHU DATASETS

Dataset		M3DCNN	HybridSN	SSRN	RNN	GCN	F2HGNNss	CEGCN	AMGCFN	MambaHSI	CSFGNet
PaviaU	Training time	37.99	21.50	25.01	1.63	43.04	48.20	74.6	185.4	465.08	93.10
	Parameters	81.88	5121.83	216.54	85.39	22.70	15.70	152.80	343.60	412.24	58.16
Xuzhou	Training time	49.40	32.62	490.70	5.23	69.10	50.70	49.13	128.20	446.02	69.20
	Parameters	337.62	5121.27	726.40	319.82	66.03	59.07	196.15	409.55	454.80	80.14
WHU	Training time	75.11	80.10	572.60	5.47	100.80	82.20	102.30	128.50	528.80	129.50
	Parameters	428.51	5122.56	471.76	375.76	45.10	39.40	175.80	409.55	434.90	71.13

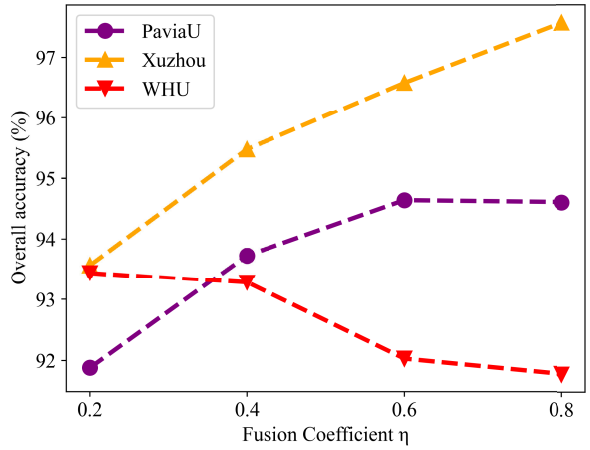


Fig. 11. OAs under different weight η values on each dataset.

for the connections of the neighborhood subgraph in the worst-case scenario. As CSGNet employs superpixels as nodes, the number of nodes is relatively small. This implies that a is typically small, resulting in negligible additional overhead. Once the transition probabilities are determined, node selection can be completed in $O(l)$. That is, generating a hyperedge of length l has a time complexity of approximately $O(l)$.

The running time (s) and the number of parameters for all the algorithms are shown in Table X. Compared to GCN and F2HGNN, CSGNet achieves a significant improvement in classification performance with only a slight increase in running time and the number of parameters. At the same time, CSGNet has a shorter running time and fewer parameters compared to the latest methods, MambaHSI and AMGCFN. This phenomenon suggests that employing hypergraphs to model high-order correlations is a promising direction worth exploring.

F. Influence of Fusion Coefficient

In the experiments, we observe that different fusion weights significantly impact the performance of the hybrid network. Therefore, in this section, we evaluate the classification performance of CSGNet on three datasets with different fusion coefficients. Specifically, the weight η for the CNN branch is set to 0.2, 0.4, 0.6, and 0.8, while the corresponding weight for the HGCN branch, 1-η, is set to 0.8, 0.6, 0.4, and 0.2. The experimental results are shown in Fig. 11. From the results we observe that on the PaviaU dataset and Xuzhou dataset, the classification performance improves when η is bigger. And on WHU dataset, the performance is higher with smaller η. This is mainly due to the fact that the PaviaU dataset and Xuzhou dataset contain a more complex distribution of land covers, and the land cover areas with the same label are small. Consequently, the pixel purity inside the superpixel block is low, leading to greater differences among similar nodes. In such cases, a higher weight for CNN can enhance the classification performance. On the contrary, on the WHU dataset, where the distribution is more concentrated and regions with the same label are larger, a higher HGCN weight leads to better performance.

V. CONCLUSION

In this article, we focus on mining the high-order correlations in HSI and propose a CSGNet for HSI classification. CSGNet employs kernel function to map the original spectral–spatial features into a high-dimensional feature space, yielding a more accurate node metric relationship and improving classification within classes. Then, CSGNet models the spatial distribution structure of features through hypergraphs (i.e., nodes with similar distribution structure are connected by a single hyperedge), which provides richer contextual information for the model and thus improves the ability of edge detection. The experimental results demonstrate that CSGNet has a strong ability to capture high-order correlations, effectively expanding the application of hypergraphs in HSI classification. However, how to effectively mine the features of hypergraph representations still remains to be a challenge. Therefore, in future work, we plan to explore hypergraph convolution to better learn the high-order correlations contained within hypergraphs.

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